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Double funnelling in a mature coastal British Columbia forest: spatial patterns of stemflow after infiltration

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Abstract:

Many studies have focused on the amount of stemflow in different forests and for different rainfall events, but few studies have focused on how stemflow intensity varies during events or the infiltration of stemflow into the soil. Stemflow may lead to higher water delivery rates at the base of the tree compared with throughfall over the same area and fast and deeper infiltration of this water along roots and other preferential flow pathways. In this study, stemflow amounts and intensities were measured and blue dye experiments were conducted in a mature coniferous forest in coastal British Columbia to examine double funnelling of stemflow. Stemflow accounted for only 1% of precipitation and increased linearly with event total precipitation. Funnelling ratios ranged from less than 1 to almost 20; smaller trees had larger funnelling ratios. Stemflow intensity generally was highest for periods with high-intensity rainfall later in the event. The maximum stemflow intensities were higher than the maximum precipitation intensities. Dye tracer experiments showed that stemflow infiltrated primarily along roots and was found more frequently at depth than near the soil surface. Lateral flow of stemflow was observed above a dense clay layer for both the throughfall and stemflow experiments. Stemflow appeared to infiltrate deeper (122 cm) than throughfall (85 cm), but this difference was in part a result of site-specific differences in maximum soil depth. However, the observed high stemflow intensities combined with preferential flow of stemflow may lead to enhanced subsurface stormflow. This suggests that even though stemflow is only a very minor component of the water balance, it may still significantly affect soil moisture, recharge, and runoff generation. Copyright © 2016 John Wiley & Sons, Ltd.

KEY WORDS stemflow; preferential flow; rainfall intensity; double funnelling; blue dye experiment; Malcolm Knapp Research Forest; mature coastal forest

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INTRODUCTION

Stemflow is often assumed to be only a small component of the water balance of mature forests, even though it can constitute up to 20% of gross incident precipitation in some forests (Levia and Frost, 2003; Johnson and Lehmann, 2006), and previous studies may have underestimated the amount of stemflow (Germer *et al.*, 2010). However, stemflow is hydrologically and biologically important because it can influence soil moisture and nutrient dynamics around trees and affect plant water uptake. Because stemflow concentrates precipitation at the base of the tree where macropores and tree roots allow stemflow to infiltrate deeper and faster into the soil (Johnson and Lehmann, 2006; Liang *et al.*, 2007, 2011), it can also contribute to a disproportionately large amount of groundwater recharge (Taniguchi *et al.*, 1996; Chang and Matzner, 2000; Liang *et al.*, 2007; Li *et al.*, 2008) and soil water recharge or saturation around the base of the tree (Germer, 2013; Buttle *et al.*, 2014). Most previous studies have focused on the amount of stemflow in different forests and climatic conditions or the amount of precipitation required for the initiation of stemflow (Aboal *et al.*, 1999; Germer *et al.*, 2010; Carlyle-Moses and Schooling, 2015), but this double funnelling of trees (i.e. the combination of above-ground funnelling of precipitation to stemflow and the below-ground funnelling of infiltrating stemflow in the soil via macropores around roots; Johnson and Lehmann, 2006; Schwärzel *et al.*, 2012) has received considerably less testing.

The various aspects of stemflow generation were reviewed in Levia and Frost (2003) and Levia and Germer (2015). The main factors that influence stemflow are tree morphology, such as leaf shape, branch structure and angle, bark texture and water holding capacity, epiphyte

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cover, and climatic characteristics, such as wind direction, and precipitation duration, timing, and intensity. The individual effects of these factors on stemflow production are difficult to separate because of their complex interaction (Levia and Germer, 2015). Levia *et al.* (2015) analysed the influence of these factors on individual European beech (*Fagus sylvatica* L.) saplings and determined that the woody surface area (biomass and branch number) per unit projected crown area had the largest influence on the quantity of stemflow produced. Tree size [diameter at breast height (DBH) or basal area] also affects the amount of stemflow (Germer *et al.*, 2010), but its effect on stemflow depends on the tree species. In general, funnelling ratios are higher for small trees than for large trees in tropical forests (Germer *et al.*, 2010), pine plantations (Buttle *et al.*, 2014), and deciduous maritime forests (Levia *et al.*, 2010; Siegert and Levia, 2014). However, few studies have been conducted in coastal coniferous forests (Rothacher, 1963; Spittlehouse, 1998; Iroumé and Huber, 2002).

After initial wetting, stemflow generally increases linearly with increasing precipitation (Manfroi *et al.*, 2004; Li *et al.*, 2008). The amount of precipitation required to wet the tree surface varies widely between species (Aboal *et al.*, 1999) and depends on antecedent wetness conditions (Xiao *et al.*, 2000; Siegert and Levia, 2014). Higher precipitation intensities can result in larger amounts of stemflow (Van Elewijck, 1989; Aboal *et al.*, 1999). Channelization of stemflow into bark furrows (Levia and Herwitz, 2005) and along preferred pathways (Liang *et al.*, 2011; Levia and Germer, 2015) after the bark water-holding capacity is filled may enhance stemflow rates compared with precipitation rates. However, Levia *et al.* (2010) showed that stemflow funnelling ratios decreased as the 5-min precipitation intensity increased, and suggested that this is a result of the conversion of stemflow to drip and throughfall when the maximum transport capacity of stemflow is exceeded. The timing of high-intensity precipitation periods during a rainfall event also affects stemflow. Dunkerley (2014) simulated rainfall on an artificial olive tree in a laboratory and determined that stemflow amounts were larger for events where the peak precipitation intensity occurred later in the event than for events where the peak precipitation intensity occurred early in the event. Still relatively little is known about how stemflow rates vary with precipitation rates in mature forests.

Because stemflow is a concentrated source of water and solutes to the forest floor, it influences groundwater recharge (Taniguchi *et al.*, 1996), soil water recharge (Germer, 2013; Buttle *et al.*, 2014), soil pH, soil nutrient status (Chang and Matzner, 2000; Hamdan and Schmidt, 2012), and soil moisture dynamics (Li *et al.*, 2008; Liang *et al.*, 2011). Despite this growing acknowledgement of the role of stemflow infiltration on soil moisture dynamics and groundwater recharge, only a few studies have attempted to observe preferential flow of stemflow (Martinez-Meza and Whitford, 1996; Li *et al.*, 2009; Liang *et al.*, 2011; Schwärzel *et al.*, 2012). Often, these studies have been conducted on desert or semi-arid shrubs (e.g. Martinez-Meza and Whitford, 1996; Li *et al.*, 2009), with the exception of Liang *et al.* (2011) and Schwärzel *et al.* (2012). Liang *et al.* (2011) mimicked throughfall and stemflow on a Stewartia tree (*Stewartia monadelpha*) on a hillslope in central Japan using two dye tracers. Schwärzel *et al.* (2012) simulated stemflow on one European Beech (*F. sylvatica*) tree using a sprinkler hose wrapped around the tree. Both studies showed that stemflow was able to bypass the soil matrix along root channels and was laterally redistributed to the downslope side of the tree (Liang *et al.*, 2011; Schwärzel *et al.*, 2012). Liang *et al.* (2011) found that stemflow infiltrated deeper in the soil than throughfall, but Schwärzel *et al.* (2012) found that a soil layer of low hydraulic conductivity impeded infiltration of stemflow into deeper soil layers.

Although these studies have begun to examine preferential flow of stemflow, more information is needed, particularly in humid coniferous forests, where stemflow is not a significant portion of the overall water balance, but preferential flow of stemflow may lead to saturation above lower permeability layers and enhanced lateral subsurface flow. This is especially the case for events for which the stemflow intensity at the base of the tree (stemflow volume per unit basal area per unit time) is larger than the rainfall intensity. The aim of this study was, therefore, to examine the double funnelling of stemflow in a mature Western hemlock (*Tsuga heterophylla*)–Western red cedar (*Thuja plicata*) forest in British Columbia (BC), Canada. More specifically, we determined (i) how much above-ground funnelling occurred and how stemflow varied with tree size and tree species; (ii) how periods of higher intensity rainfall during an event affect the stemflow intensity; (iii) the pathways of infiltrating stemflow in the soil; and (iv) if stemflow infiltrates to a greater depth than throughfall.

STUDY SITE DESCRIPTION

The 0.9 ha study site is located near the middle of the 5157 ha Malcolm Knapp Research Forest (MKRF), approximately 60 km east of Vancouver, BC (49°17′N, 122°36′W; Figure 1). The MKRF is located within the Coastal Western Hemlock biogeoclimatic zone. This region has a maritime climate. Annual precipitation ranges from approximately 2200 mm/y at the southern end of the MKRF to 3000 mm/y at the northern end of

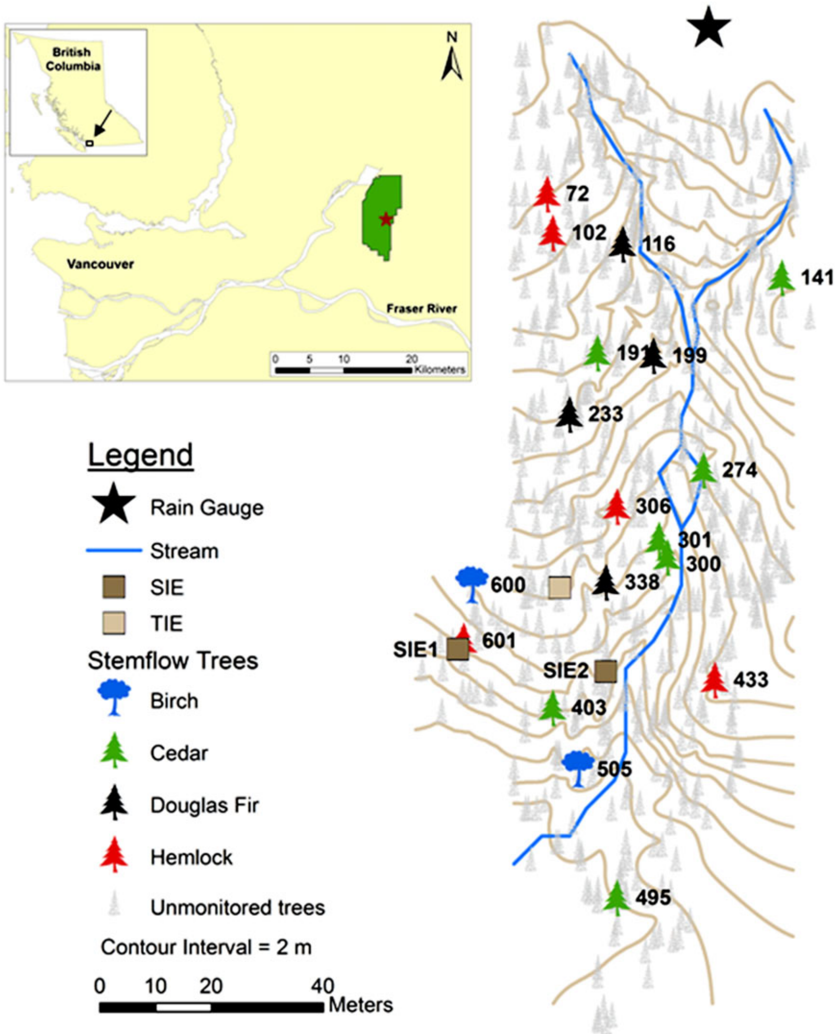

Figure 1. Location of the trees for which stemflow was measured and the locations of the blue dye experiments in the study site. The insert shows the location of the Malcolm Knapp Research Forest (green polygon) in southwestern British Columbia and the study site (star) within the Malcolm Knapp Research Forest. The numbers represent the tree numbers; refer to Table I for the details of the measurement period for each monitored tree

the research forest (MKRF, 2010a, b). Seventy per cent of the precipitation falls between October and March (primarily in the form of rain). Soils at the study site are Ferro-humic Podzols and Brunisols on glacial till and granodiorite bedrock (Wang *et al.*, 2002; Haught and van Meerveld, 2011). Average soil depth is less than 1 m.

The study site was logged in the 1920s and burned in a large forest fire in 1931. The forest regenerated naturally afterwards; the current forest is approximately 80 years old (MKRF, 2010b). The forest at the study site is a mixed-species stand with approximately 600 trees/ha and a basal area of 57 m^2/ha. Western red cedar (*T. plicata*; 42% of all trees) and Western hemlock (*T. heterophylla*; 41% of all trees) are the primary tree species, although birch (*Betulaceae spp.*; 7%), Douglas-fir (*Pseudotsuga menziesii*; 7%), and vine maple (*Acer circinatum*; 3%) are also present in the study site. Average (and median) DBHs are 31 cm (27 cm), 30 cm (30 cm), 22 cm (19 cm), 38 cm (32 cm), and 5 cm (5 cm) for the cedar, hemlock, birch, Douglas-fir and maple trees respectively.

METHODS

Above-ground funnelling

Stemflow measurements. Stemflow was collected from five Western hemlock (*T. heterophylla*), seven Western red cedar (*T. plicata*), four Douglas-fir (*P. menziesii*), and two Birch (*Betulaceae spp.*) trees of different sizes from February 2010 to May 2011 (Table I and Figure 1). Stemflow from maple trees was not measured because there are few maple trees within the study site (only 3% of all trees, representing 0.1% of the basal area), and their small complex branches make stemflow measurements difficult. Soft plastic tubing (2 cm inside diameter, with one third cut out to create a trough) was nailed to the trees

Table I. Details of the stemflow measurements.

Tree number	Species	DBH (cm)	*Measurement period	
			Collecting bucket	Tipping bucket
141	Cedar	37	12 Feb 10–15 Nov 10	–
191	Cedar	16	25 Jun 10–5 Nov 10	–
274	Cedar	32	27 May 10–15 Nov 10	–
300	Cedar	57	12 Feb 10–5 Nov 10	–
301	Cedar	30	12 Feb 10–26 Mar 10	17 Apr 10–7 Mar 11
403	Cedar	31	12 Feb 10–5 Nov 10	5 Nov 10–4 May 11
495	Cedar	23	25 Jun 10–15 Nov 10	–
72	Hemlock	31	12 Feb 10–15 Nov 10	2 Dec 10–7 Mar 11
102	Hemlock	41	28 Jun 10–15 Nov 10	–
306	Hemlock	50	12 Feb 10–15 Nov 10	–
433	Hemlock	30	12 Feb 10–15 Nov 10	2 Dec 10–7 Mar 11
601	Hemlock	29	–	25 Jun 10–4 May 11
116	Douglas-fir	43	12 Feb 10–15 Nov 10	2 Dec 10–7 Mar 11
199	Douglas-fir	26	25 Jun 10–15 Nov 10	–
233	Douglas-fir	34	27 May 10–5 Nov 10	5 Nov 10–4 May 11
338	Douglas-fir	31	12 Feb 10–15 Nov 10	–
505	Birch	27	12 Feb 10–15 Nov 10	–
600	Birch	16	27 May 10–5 Nov 10	5 Nov 10–4 May 11

*Freezing temperatures and snow resulted in only a few useful events between mid-November 2010 and February 2011 for all tipping buckets. For the location of the trees, refer to Figure 1.

in a spiral orientation (at least 1.5 rotations around the tree), starting at approximately 1.8-m above the surface to direct stemflow down the tree stems. Silicone sealant and putty were used to fill holes between the tubing and the bark. Stemflow was routed to either 19l water-collecting containers with a lid or tipping bucket rain gauges. The volume of water in the containers was measured with a 1000-ml graduated cylinder after each precipitation event. Two types of tipping bucket rain gauges were used: 5-ml tipping bucket rain gauges (Onset, Cape Cod, MA, USA) for trees with a consistently small volume of stemflow per event and custom-built larger tipping bucket rain gauges with a 25-ml tipping volume (Table I).

Precipitation was measured using a tipping bucket rain gauge (0.2 mm/tip; Onset, Cape Cod, MA, USA) in a clearcut located directly north of the study site (Figure 1). Precipitation measured in the clearcut is assumed to represent the precipitation falling on the canopy. Precipitation data are missing or unreliable from mid-November 2010 to February 2011 as a result of battery malfunction, freezing temperatures, and snow. Rainfall events during this period were therefore excluded from the analyses.

Stemflow data analysis. Stemflow volumes were converted to a representative depth by dividing the measured stemflow volume per event by the basal area of the tree. Precipitation events were defined as periods with more than 2.5 mm of precipitation, separated by at least 12 h of no precipitation. The stemflow data were divided into events based on the assumption that stemflow can only begin after the start of a rainfall event. Because stemflow often continued after rainfall ended, the end of a stemflow event was defined either as the start of the next rainfall event or the start of a 12-h period with no stemflow. For each event, the total depth (volume per unit basal area), peak intensity, and average intensity of stemflow and precipitation were calculated at an hourly time-step. An hourly time interval was chosen because of the small stemflow volumes (and thus long time between consecutive tips) for the coniferous trees. Funnelling ratios (F) describe the degree that precipitation is concentrated as stemflow and are calculated by dividing the stemflow depth (i.e. volume per unit basal area) by the precipitation depth for each event (c.f. Herwitz, 1986).

Below-ground funnelling: blue dye staining experiments

Two stemflow and one throughfall dye staining experiments (Figure 1) were conducted in the study site using Brilliant Blue dye. Brilliant Blue dye is frequently used for dye staining experiments because of its low sorption and high mobility (allowing the dye to travel through preferential flow pathways) and because it has little interaction with the soil matrix, which results in a highly defined edge of the pathways (Anderson *et al.*, 2009). The stemflow and throughfall dye-staining experiments were located within 30 m from each other to ensure that the soil characteristics were similar. Soil depth differed slightly between the sites, but the soil layers and soil texture were similar. To ensure that the antecedent moisture conditions were relatively similar for the experiments, surface soil moisture was measured at 27 locations throughout the study site using a 20-cm HydroSense probe (Campbell Scientific) prior to the experiments. Average soil moisture was 14.2% for stemflow experiment 1 (SIE 1) in September 2010 and 13.5% for the throughfall experiment (TIE) in August 2011. Stemflow experiment 2 (SIE 2) began 10 days after the throughfall experiment. It is likely that the average soil moisture was slightly lower than 13.5% for stemflow infiltration experiment 2 because there was no precipitation between the throughfall experiment and the start of stemflow experiment 2. It is assumed that this small difference in soil moisture does not significantly affect the results.

For the first stemflow infiltration experiment (SIE 1), 18 l of Brilliant Blue dye (5 g/l) were sprayed evenly onto a Western hemlock stem (tree 601; DBH = 29 cm) over a 3-h period using a backpack sprayer. This amount of stemflow was previously measured for this tree during a 50-mm rainfall event. The soil surrounding the tree was excavated 2 days after application of the dye and analysed for the presence of dye (cf. Weiler and Naef, 2003; Anderson *et al.*, 2009) in order to determine the location and type of preferential flow pathways. The excavation began 2 m away from the base of the tree because Wang *et al.* (2002) suggested that Western hemlock tree roots in coastal BC can expand far from the tree. The width of the excavation was 3 m. Careful excavation occurred in slices 30 cm apart at 2 m from the tree and decreased to 10 cm apart as the excavation moved closer to the tree (Figure S1a). For each profile, detailed notes and digital photographs were taken, and scaled diagrams were drawn to document the location and pattern of the preferential flow pathways and the location and distribution of roots, rocks, and boulders. Tarps were hung above the plot and around the tree to avoid dilution of the dye during the 5-week excavation period. However, subsurface stormflow at the soil-bedrock interface may have influenced the dye in the lowest 5–10 cm of the soil profile.

For the second stemflow infiltration experiment (SIE 2), 6 l of diluted Brilliant Blue dye (5 g/l) were sprayed on another Western hemlock tree (tree 411; DBH = 52 cm) over a 2.5-h period. This amount represented the amount of stemflow that was measured during a 50-mm event for a similar sized Western hemlock tree (tree 306; DBH = 50 cm). The excavation began only 1 m from the tree and occurred in slices 20 cm apart, decreasing to 10 cm apart as the excavation moved closer to the tree (Figure S1b). Excavation for SIE 2 was completed in 4 weeks.

For the TIE, 40 l of diluted Brilliant Blue dye were sprayed evenly over a 3-h period on a 0.81 m^2 area that was free of large shrubs and trees (representing 50 mm of throughfall or approximately 60 mm of precipitation). The excavation began 10 cm into the sprayed plot to avoid edge effects; excavation slices were 10–20 cm apart (Figure S1c).

Direct analysis of the photos, as carried out in other dye-staining studies (e.g. Weiler and Fluhler, 2004; Bachmair *et al.*, 2009; Bogner *et al.*, 2010; Schneider *et al.*, 2014), was not possible because the large roots (up to 40 cm in diameter) and large boulders (more than 1 m in diameter) caused significant shadows in the photographs and restricted the view of the camera (Figure S2). This resulted in photographs that only covered parts of the excavated area, photographs that were stretched in different ways, or poor quality photographs with many shadows, thus difficulty in automatic processing of the images. Instead, the detailed notes, drawings, and photos were used to digitize the location of all blue dye stains, as well as all roots and rocks larger than 2 cm in diameter in Google Sketchup®.

To determine the fraction of soil that contained blue dye, the number of blue squares in the scaled drawings (representing a 10 × 10 cm area in the soil profile) were counted and divided by the total number of squares that represented the width of that subsection. The widths of the profiles up to the tree edge (located at $x = 0$ m) were 160 cm for SIE 1, 200 cm for SIE 2, and 90 cm for TIE. The widths decreased beyond the tree edge for SIE 1 and SIE 2 because these profiles only represented the downslope part of the excavation.

RESULTS

Stemflow amount

Stemflow increased linearly with precipitation once wetting had occurred (average coefficient of determination: 0.85; Figure 2). The minimum amount of precipitation required for stemflow to occur (the intercept of the *x*-axis in Figure 2) varied between 1 and 14 mm (average 8 mm) and did not depend on species or tree size. Funnelling ratios (F) increased with event size until a maximum and remained constant thereafter (Figure 3). This plateau occurred, on average, after approximately 50 mm of precipitation. For some trees, the funnelling ratios decreased for events larger than 100 mm (e.g. trees 403 and 233; Figure 3). The relation between funnelling ratios and DBH shows that the smaller trees were able to funnel more stemflow per unit basal area than the larger trees (Figure 4). For the larger trees, the funnelling ratios were smaller than 1 and did not depend on event size. Event size had a much larger effect on the funnelling ratios of small trees (refer to the larger range of data points for the smaller trees in Figure 4); the funnelling ratios were generally larger for larger events, but there was considerable scatter in the relation.

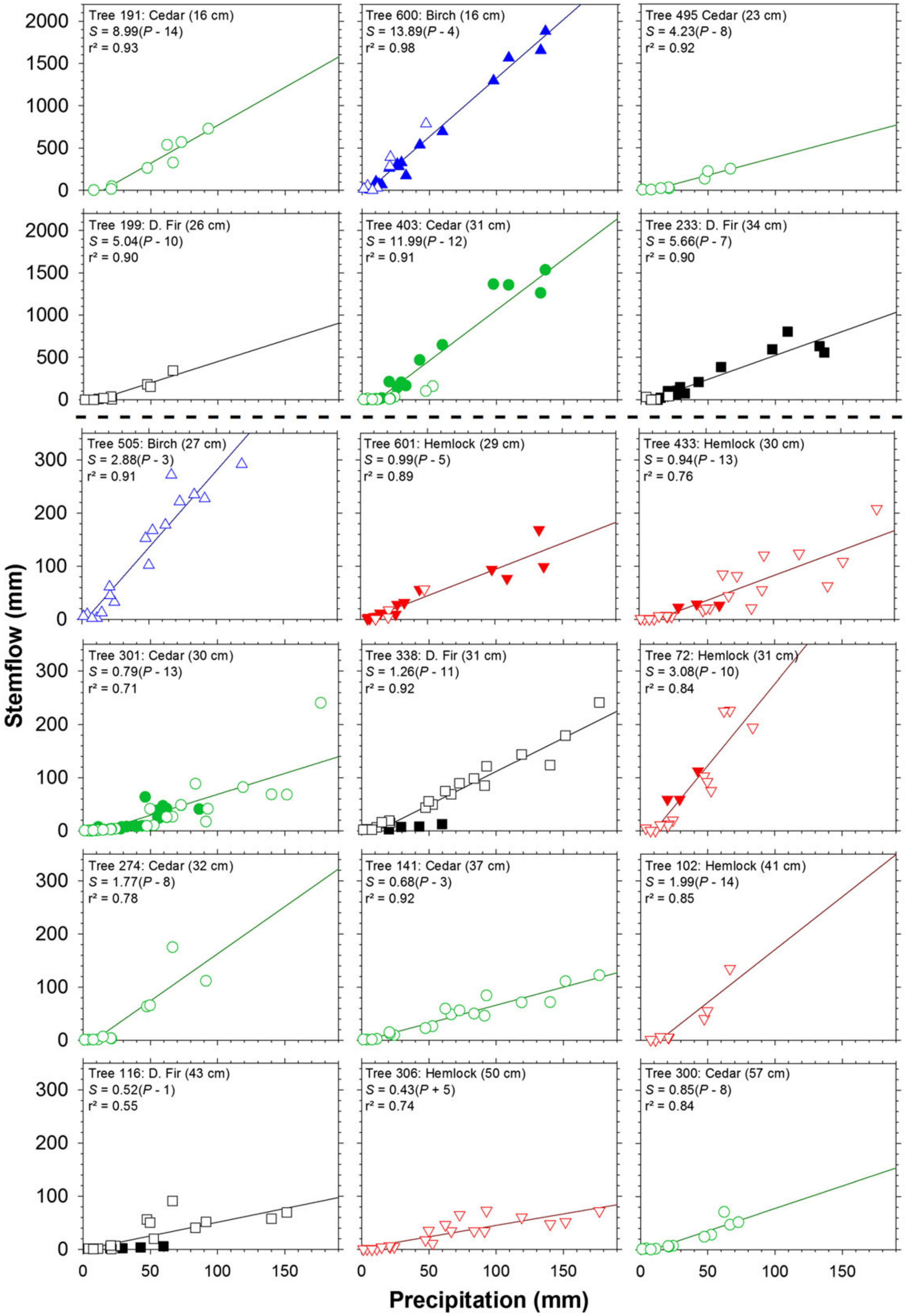

Figure 2. Stemflow per unit basal area (S) as a function of precipitation (P) for all studied trees. Open symbols represent data from the water-collecting containers; closed symbols represent data from the tipping buckets. All regression lines are statistically significant ($p < 0.05$). Tree number, species, and diameter at breast height (DBH) are given in the top left corner of each subplot. For the location of the trees, refer to Figure 1. For the measurement period, refer to Table I. Note the change in axes below the second row of subplots (indicated by the dashed line). Within each group, the data are organized according to tree size (smallest tree top left to largest tree bottom right)

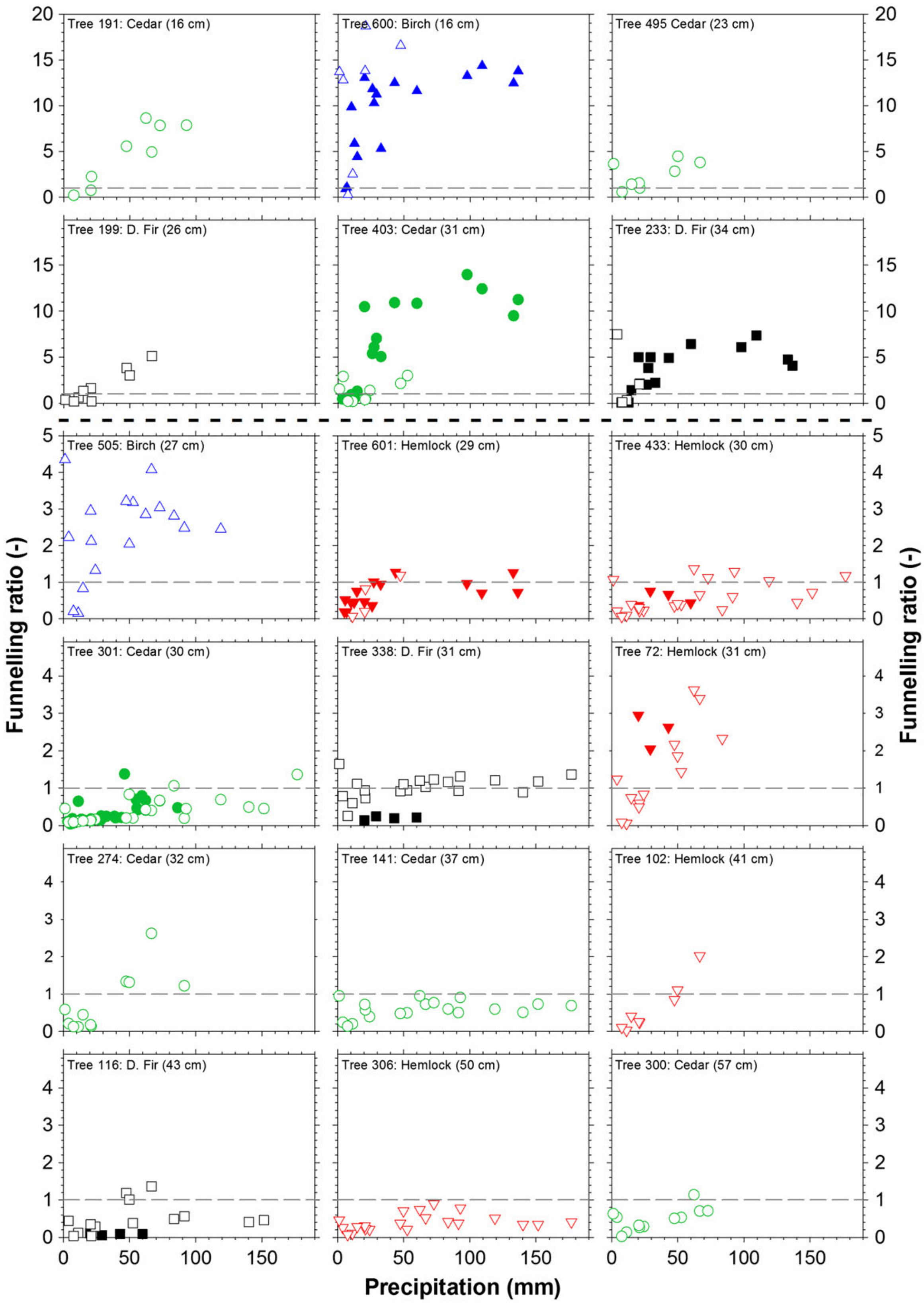

Figure 3. Funnelling ratios (*F*; unitless) as a function of event size for all monitored trees. Open symbols represent data from the water-collecting containers; closed symbols represent data from the tipping buckets. DBH is given in parentheses. The grey dashed line represents a funnelling ratio of 1 (i.e. stemflow per unit basal area equals precipitation). The subplots are organized in the same order as in Figure 2

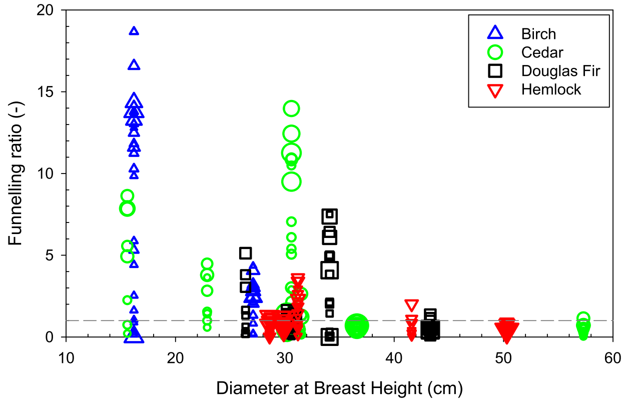

Figure 4. Funnelling ratio (F) as a function of diameter at DBH for all species and events. The horizontal grey line represents a funnelling ratio of 1. Symbol size reflects the event size, with larger symbols representing larger events

Total stemflow from all trees in the watershed was estimated based on the linear relations between stemflow and precipitation [$S = a(P - b)$], where S is the stemflow amount (mm), P is precipitation (mm), a is the slope of the relation between stemflow and precipitation, and b is the minimum precipitation before stemflow occurs (Figure 2). The slope of the relation between stemflow and precipitation (a) depended on DBH (Figure S3). The minimum amount of precipitation required before stemflow occurred (b) was set at 8 mm for all trees, as this was the mean of the maximum event size for which no stemflow was measured for all studied trees. Using individual regressions for the five species or separating the data into size classes (c.f. Germer *et al.*, 2010) did not improve the regression, and thus, a single equation [$S = a(P - 8)$] was used to estimate the amount of stemflow for all 550 trees in the study site for all events for which precipitation was measured during the spring to winter 2010 period (total precipitation: 1311 mm). The estimated total stemflow for all trees during this study period was 13 mm or 1% of precipitation. When considering the entire study site with a total basal area of 51 m^2 (0.57% of the total study area), this leads to an overall average funnelling ratio of 2. Trees with a DBH less than 35 cm contributed 72% of the estimated total stemflow amount. These trees represent only 24% of the total basal area.

Stemflow intensity

Peak stemflow intensities increased with peak precipitation intensity (Figure 5). There may be a threshold at 2 mm/h, above which peak stemflow intensities increased faster than peak precipitation intensities, but this cannot be analysed because of the limited number of events and trees for which stemflow intensity data are available. For some trees, peak stemflow intensities were

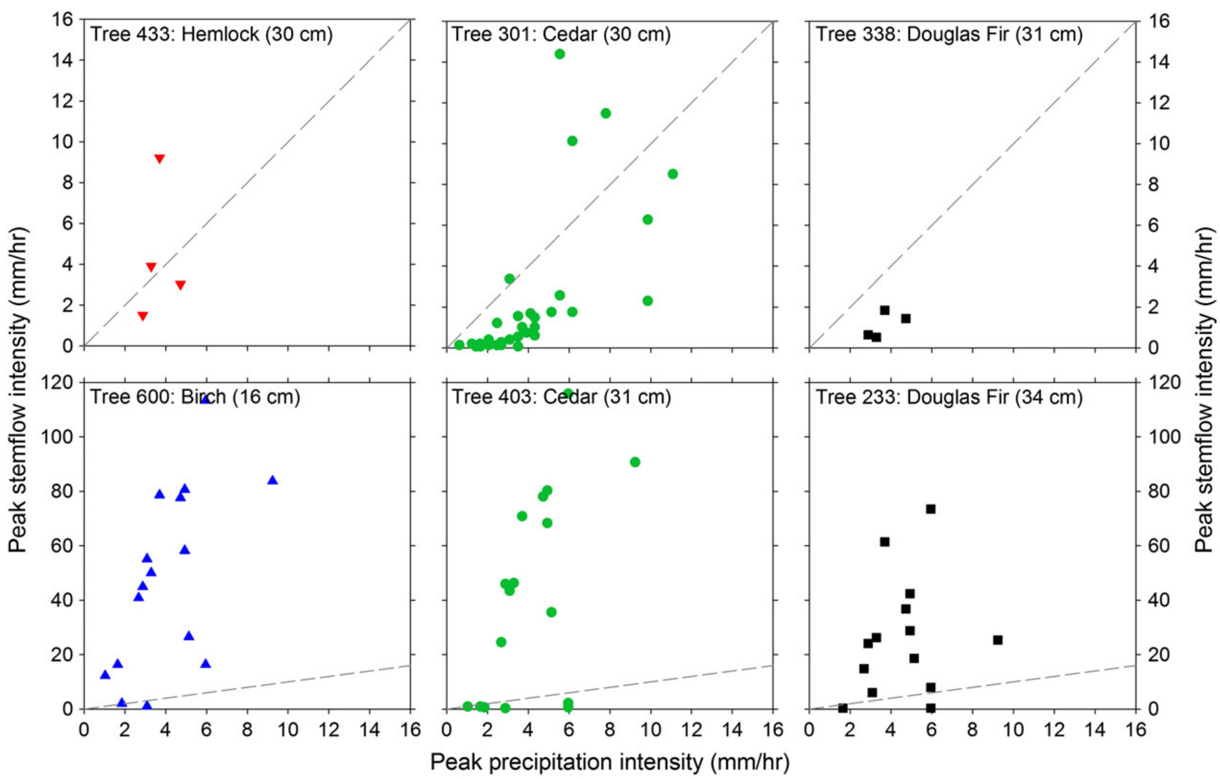

Figure 5. Peak stemflow intensity as a function of peak precipitation intensity for each event. The dashed line represents the 1 : 1 line. DBH is given in parentheses

much higher than peak precipitation intensities (e.g. trees 600, 403, and 233). For other trees, peak stemflow intensities were predominantly lower than the peak precipitation intensities but were higher than the peak precipitation intensity for some events (e.g. trees 433 and 301). A comparison of the precipitation and stemflow time series (Figures 6 and 7) shows that for some events, stemflow peak intensities increased in a stepwise manner, regardless of the precipitation intensity (e.g. for tree 403 and 233; Figure 6d). When there were two periods of higher precipitation intensity, stemflow often peaked during the second period of higher precipitation intensity. However, when there were several hours between consecutive periods of higher precipitation intensity, there was often a lower stemflow intensity during the next period of higher precipitation intensity (Figure 6c). As a result, the maximum stemflow intensity usually occurred later in the event and not at the same time as the maximum precipitation intensity (Figures 6 and 7).

Blue dye infiltration experiments

In SIE 1, blue dye was observed on live and dead roots, inside dead roots, around rocks, and in the soil matrix. The maximum observed depth of the dye was 122 cm, approximately 9 cm above the bedrock (Figure 8). Laterally, blue dye stains were confined within 50 cm of the tree, with the exception of one location at approximately 1 m from the tree where blue dye was found at 55 to 65 cm below the soil surface. The location and pattern of the dye stains suggest that stemflow preferentially flowed to the downslope side of the tree (Figure 8) and flowed primarily through the 10-cm organic rich top layer of the soil around the tree before flowing between or along roots (and rocks) deeper into the soil. There was a strong bimodal distribution of blue dye stains with depth, with a higher occurrence of blue dye at approximately 20 and 70 cm below the soil surface and a higher prominence at 70 cm than at 20 cm (Figure 9). A dense clay layer was observed at 70 cm below the soil

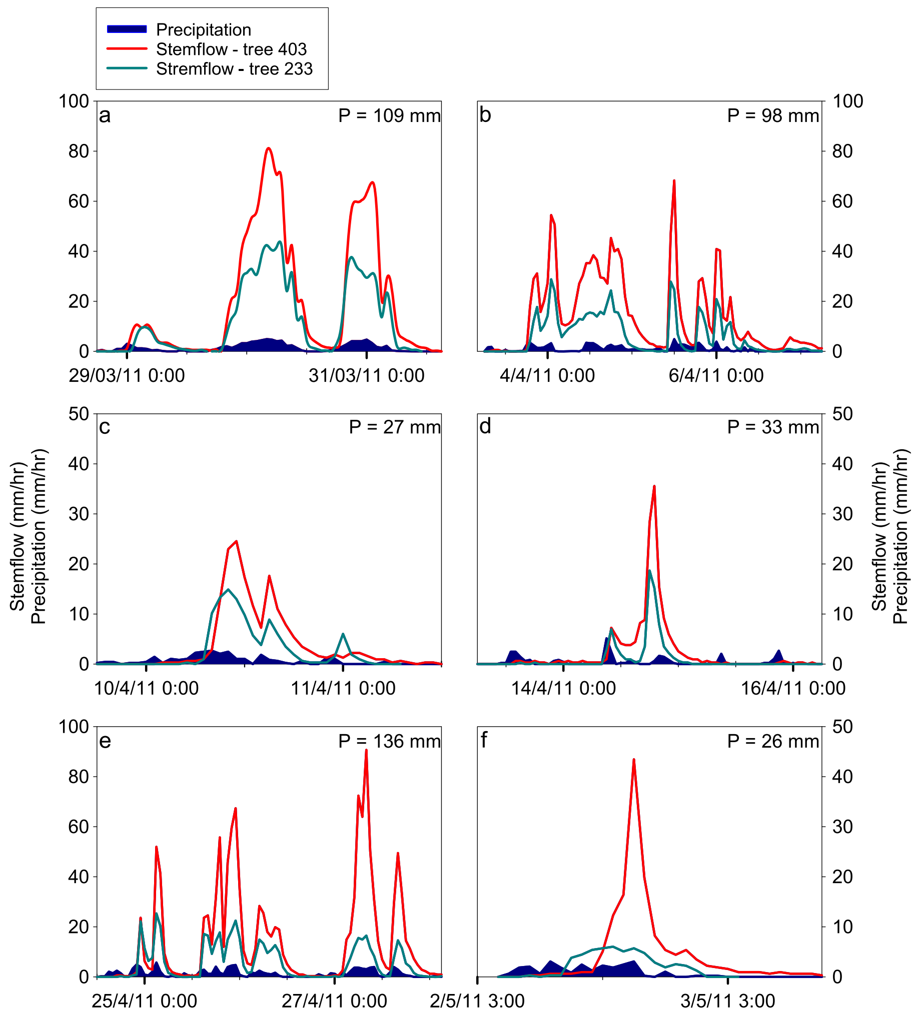

Figure 6. Precipitation and stemflow intensity time series for six events for tree 403 (cedar; 31 cm DBH) and tree 233 (Douglas-fir; 34 cm DBH). Total event precipitation is reported in the top right corner

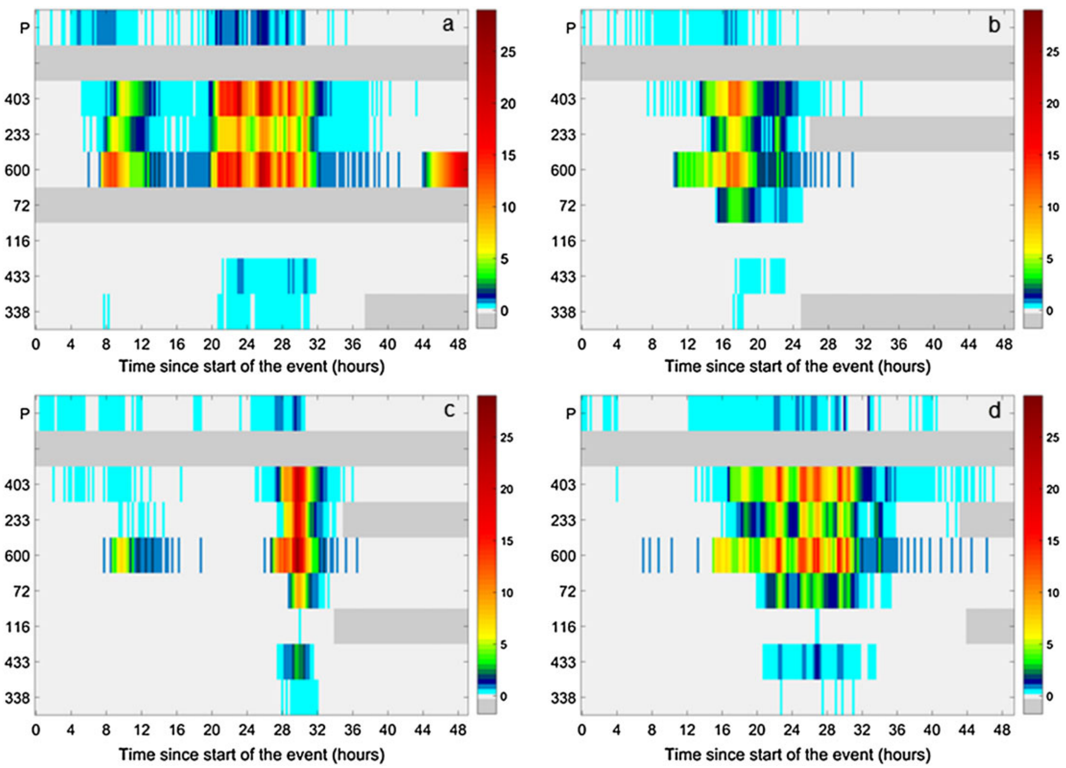

Figure 7. 15-min precipitation intensity (*P*; top row of each subplot) and stemflow intensity for seven trees (both in mm/15 min) as a function of time since the start of the event for the 60-mm event on 3–4 Feb. 2011 (a), the 20-mm event on 6 Feb. 2011 (b), the 39-mm event on 11–12 Feb. 2011 (c), and the 41-mm event on 13–15 Feb. 2011 (d). Missing data are indicated by dark grey shading

surface. At the downslope side of the tree (30 cm beyond the tree edge), the stains no longer showed the bimodal distribution but were very prominent at 50 to 90 cm below the soil surface (Figure 8d), suggesting that stemflow bypassed the top 50 cm of soil.

For SIE 2, blue dye stains were confined within 60 cm of the tree (Figure 10). Blue dye was observed on live and dead roots, inside dead roots, around rocks, and in the soil matrix, similar to SIE 1. The dye was more prevalent along the roots than in the soil matrix. Upslope and downslope differences were minimal for this tree. Blue dye was found frequently at 40 and 70 cm below the soil surface, above dense clay layers, but was also prevalent at the soil surface (Figures 9 and 10). These blue dye stains suggest that stemflow infiltrated through the top 10 cm of the organic rich top layer and then travelled along roots, and occasionally rocks and boulders, to deeper soil layers. The maximum infiltration depth was 77 cm below the soil surface, approximately 14 cm above the bedrock (Figure 10e).

In the throughfall infiltration experiment, the blue dye was mainly observed in the soil matrix, but it was also found around rocks and roots and in dead roots, resulting in a steady decline in blue dye occurrence with depth below the soil surface (Figures 11 and 12). Lateral flow of dye above a dense clay layer was observed for slices 4 and 6 at 40 cm below the soil surface. The maximum infiltration depth was 85 cm below the soil surface, approximately 6 cm above the bedrock. Although blue dye was sprayed evenly across the soil surface during application, there was significant variation in the blue dye patterns between the excavated profiles (Figure 12). The upslope profiles had less dye at depth than the downslope profiles. This could be a result of channelling of throughfall in the litter layer (thatched roof effect; Ward and Robinson, 2000) and the near-surface layers. Blue dye also appeared to flow preferentially to the west side of the plot. There were remnants of a decaying fallen tree and large rocks in the upper layers of the soil, particularly on the east side of the plot, which could decrease the flow into deeper soil layers and divert the flow around them.

DISCUSSION

Stemflow amount

Total stemflow was only 1% of precipitation, which is on the lower end of the range reported for temperate conifer forests (Spittlehouse, 1998; Iroumé and Huber, 2002; Levia and Frost, 2003). However, none of the studies reported in Levia and Frost (2003) were conducted in cedar-hemlock forests or along the west coast of North America, and only some were conducted in

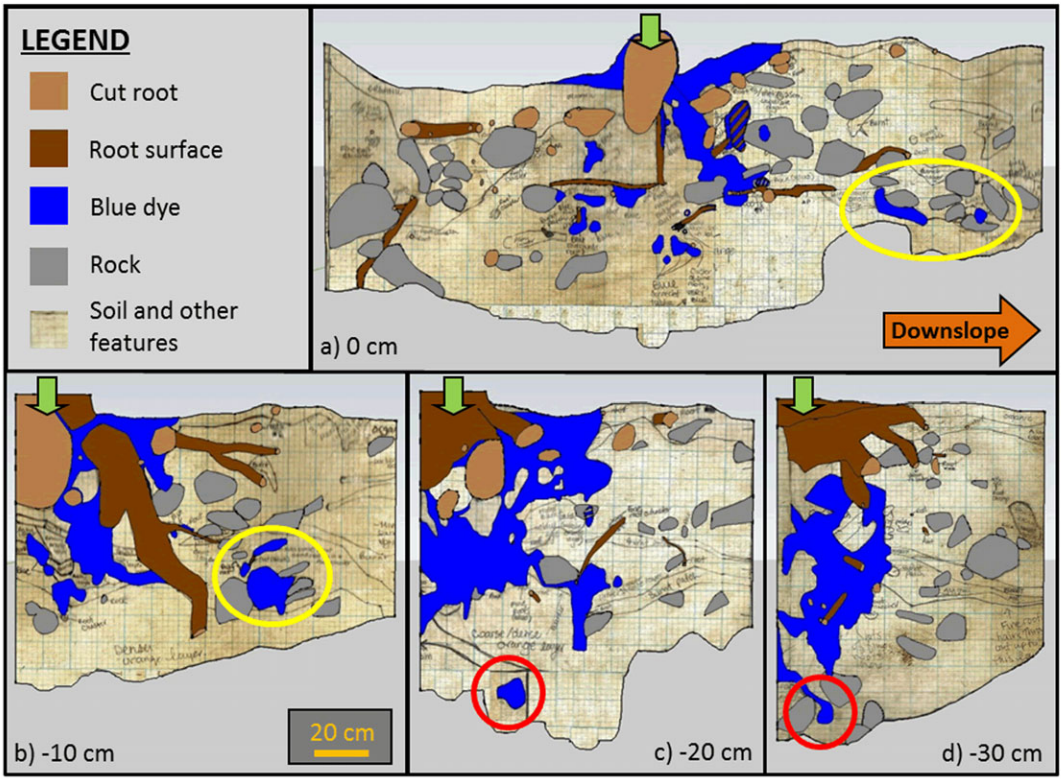

Figure 8. Diagrams showing the distribution of blue dye, roots, and large stones for SIE 1. Yellow circles in (a) and (b) indicate where stemflow flowed laterally downslope beyond 50 cm from the tree. Red circles in (c) and (d) indicate the maximum stemflow infiltration depth at 122 cm below the surface. Distances are given relative to the edge of the tree (distance of 0 cm), with negative numbers representing excavation profiles to the east. The tree stem is indicated by the green arrow. The orange arrow indicates the direction of the hillslope and is the same for all subplots

mature forests. Spittlehouse (1998) measured stemflow on Douglas-fir trees in a mature forest on Vancouver Island and also determined that stemflow was only 1% of precipitation. The overall funnelling ratio of 2 suggests, on average, only a minor enrichment of precipitation at the base of the trees. Trees with a DBH less than 35 cm (and generally $F > 1$) contributed nearly three quarters of the total stemflow, even though they only represent one quarter of the total basal area. This confirms the results of other recent studies that small trees have more stemflow per unit area (Germer *et al.*, 2010; Fan *et al.*, 2015; Honda *et al.*, 2015) so that stemflow may have a significant impact on the soil moisture dynamics around small trees. Manfroi *et al.* (2004) suggested that for a tropical rainforest, the effect of tree size on stemflow volume depends the size of the event, with smaller understory trees funnelling most water during small events and larger trees funnelling most water during large events. This was not observed for the temperate rainforest studied here.

There has been recent discussion about how tree bark affects the amount of stemflow transmitted to the tree bole (Levia and Herwitz, 2005; Van Stan and Levia, 2010; Siegert and Levia, 2014). Trees with thicker rough bark would have a larger bark water holding capacity than smooth bark trees, which results in less stemflow (Van Stan and Levia, 2010). In this study site, there are three rough bark tree species and one smooth bark tree species. Of the rough bark trees, western hemlock has overlapping scales of bark, which can create more drip points and less stemflow (Crockford and Richardson, 2000). Douglas-fir trees have thick bark with deep linear furrows, which can produce more stemflow if the preferred pathway is already wet, and stemflow is routed within these deep furrows (Levia and Herwitz, 2005). Western red cedar trees have thinner spongy bark with long overlapping strips of bark, which can create more drip points, a larger bark water holding capacity and less stemflow. However, despite these differences in bark, stemflow volumes and funnelling ratios did not vary consistently with tree species. Rather, stemflow varied more with tree size (and event size) than with tree species.

Funnelling ratios increased with event size until a threshold or maximum and remained constant thereafter (Figure 3). This plateau occurred on average at approximately 50 mm of precipitation. Carlyle-Moses and Price (2006) suggested that this threshold is likely the precipitation depth required for trees to transmit stemflow at their maximum efficiency. Stemflow occurred during events smaller than 50 mm because some flow pathways were activated. We often observed

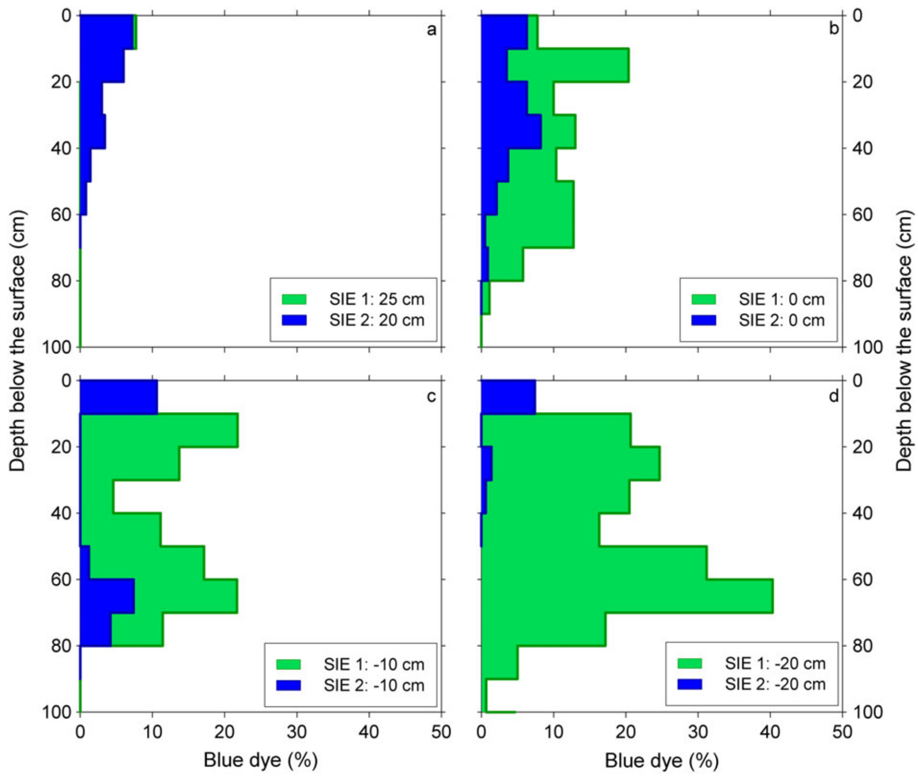

Figure 9. Percent blue dye-stained area as a function of depth below the soil surface for SIE 1 (green) and SIE 2 (blue) at 20–25 cm west of the stem (a), at the tree stem (b), and at 10 cm (c) and 20 cm (d) east of the stem

that stemflow occurred on only part of the tree stem, and that other parts of the tree stem remained dry. During larger events, more of the tree surface was wet, and flow pathways likely connected, which creates the conditions for maximum stemflow efficiency (Carlyle-Moses and Price, 2006). For some trees, the funnelling ratios decreased for events larger than 100 mm (e.g. trees 403 and 233 in Figure 3). As suggested by Carlyle-Moses and Price (2006) and Li *et al.* (2008), a tree has a maximum funnelling ability, and drip losses increase for larger and higher intensity events. However, it is difficult to determine if this trend occurred for all trees because data for events larger than 100 mm are missing for some trees as a result of the limited size of the collecting containers. The decrease in funnelling ratios after 100 mm may also be as a result of the longer duration of the large events. Most large events occurred over several days, and precipitation stopped periodically during the event (for periods shorter than 12 h so that they were not characterized as a new event) or had a low precipitation intensity for several hours.

Stemflow intensity

Peak stemflow intensities were significantly higher than the rainfall intensity for some events. This was not limited to a particular species and was also observed for trees that had overall low stemflow volumes and intensities. The stemflow intensity time series reveal the importance of antecedent precipitation on stemflow intensity and timing (c.f. Carlyle-Moses and Price, 2006; Germer *et al.*, 2010; Siegert and Levia, 2014). Stepwise increases in stemflow intensity during subsequent periods with higher rainfall intensity likely indicate that the tree reached its storage capacity and became more efficient in transmitting stemflow during the event (Carlyle-Moses and Price, 2006; Liang *et al.*, 2009). However, when there were several hours between periods of higher precipitation intensity, rewetting of the canopy and bark resulted in a decrease in stemflow intensity during the next period with higher precipitation intensity. The larger stemflow intensities later in the event are in agreement with the experiments on an artificial olive tree by Dunkerley (2014), who showed that stemflow fluxes were larger for events for which the peak rainfall intensity occurred later in the event.

A (much) higher peak stemflow intensity than peak precipitation intensity, combined with preferential flow of stemflow along roots (Crabtree and Trudgill, 1985), could result in faster recharge of groundwater (Taniguchi *et al.*, 1996) or soil water (e.g. Liang *et al.*, 2011) and may contribute to the rapid delivery of stemflow to streamflow, as suggested by Germer *et al.*

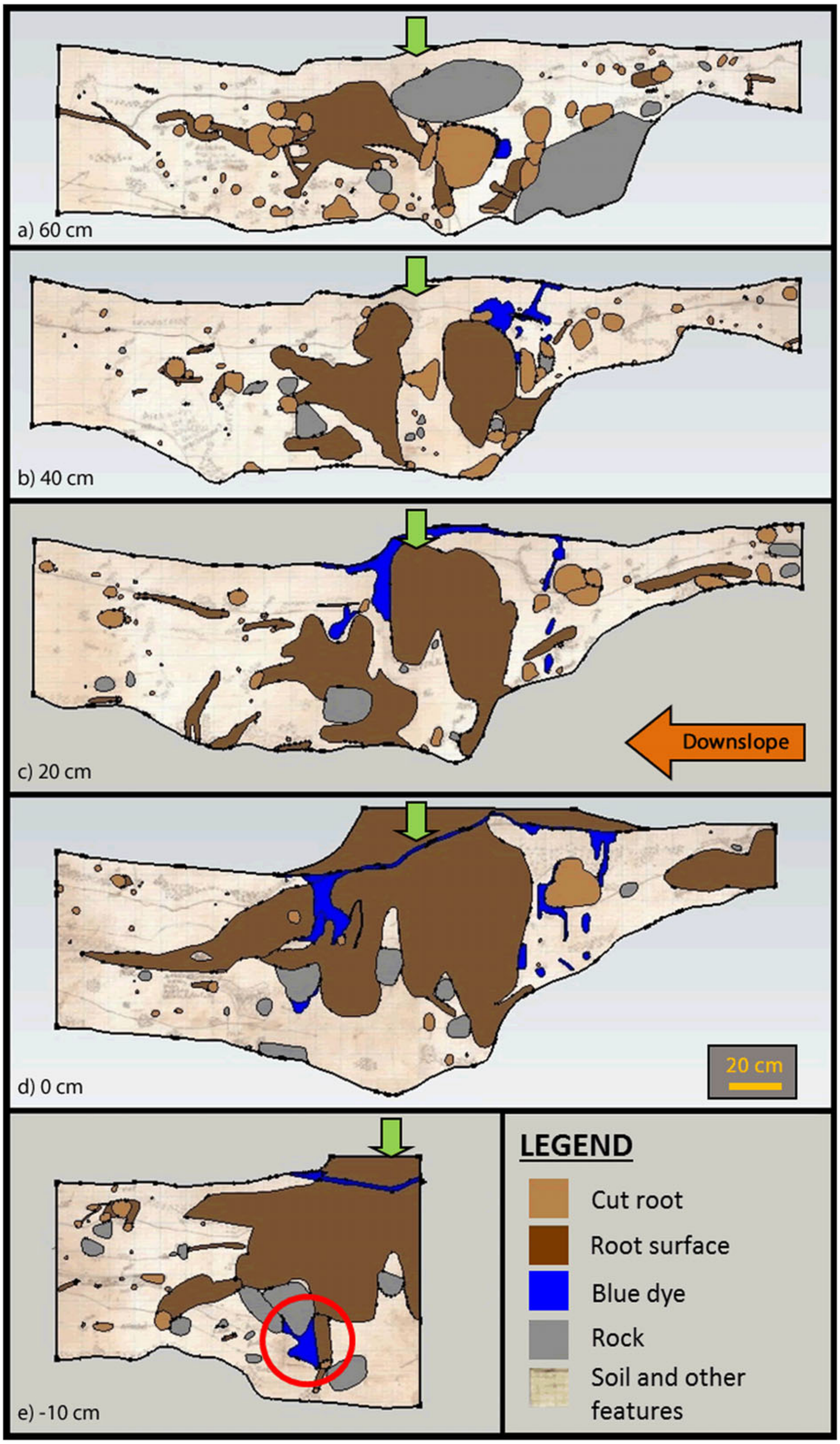

Figure 10. Diagrams showing the distribution of blue dye, roots, and large stones for SIE 2. The green arrow indicates the location of the stem. Stemflow was confined within 60 cm of the tree and generally flowed around tree roots and rocks. The red circle in figure (e) indicates the maximum stemflow infiltration depth (77 cm). Distances are given relative to the edge of the tree, with positive numbers representing excavation profiles to the north and negative numbers representing excavation profiles to the south. The orange arrow indicates the direction of the hillslope and is the same for all subplots

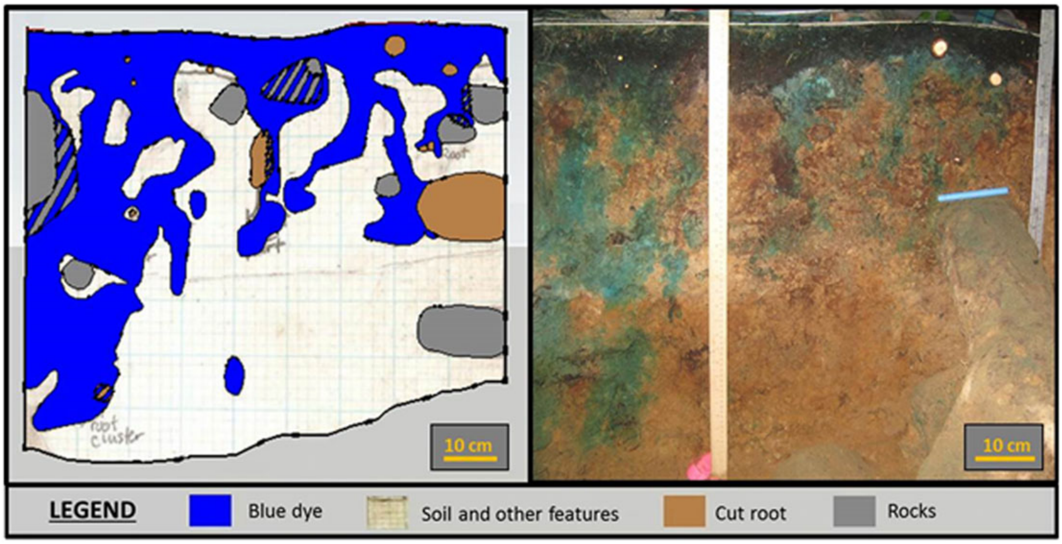

Figure 11. Diagrams showing the distribution of blue dye, roots, and large stones for the throughfall experiment

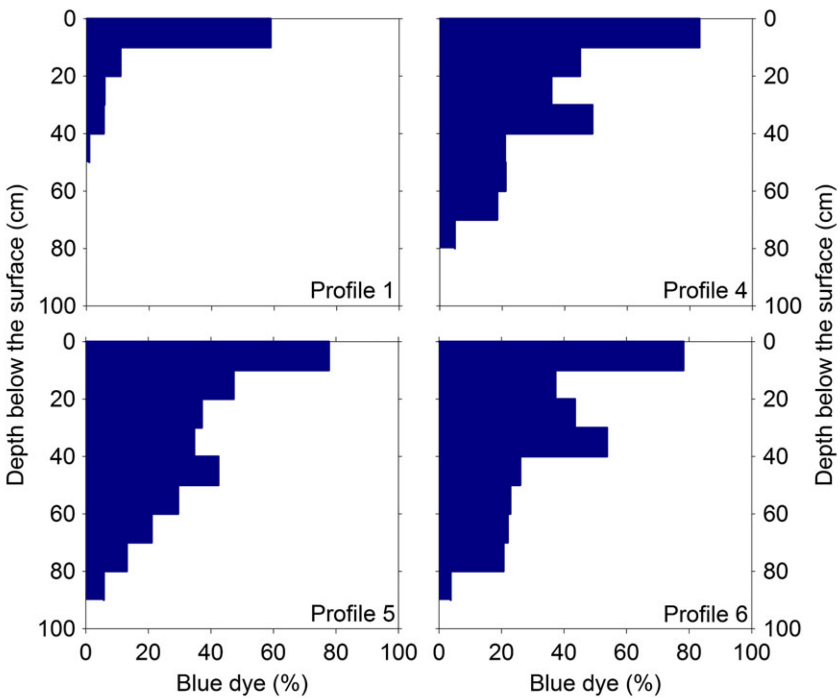

Figure 12. Percent of the cross section with blue dye as a function of depth below the soil surface for the throughfall experiment. Profile 1 is located the furthest upslope, while profile 6 is located furthest downslope. The width of the excavation was 90 cm

(2010). Previous studies on stemflow in tropical areas have suggested that stemflow intensity may be greater than the hydraulic conductivity of the soil surrounding the tree, resulting in overland flow (e.g. Herwitz, 1986; Germer *et al.*, 2010; Levia and Germer, 2015). Overland flow of stemflow is unlikely for the study site because the maximum measured stemflow intensity in this study (~120 mm/h) is still three times smaller than the hydraulic conductivity of the soil surface (Haught and van Meerveld, 2011). However, the lower soil hydraulic conductivity of the deeper soil layers could lead to lateral subsurface flow and contribute to the rapid delivery of precipitation to streamflow via subsurface stormflow rather than groundwater recharge (Herwitz, 1986; Germer *et al.*, 2010; Schwärzel *et al.*, 2012). The occurrence of many blue dye stains above clay layers also suggests enhanced lateral flow of stemflow.

Infiltration of stemflow into the soil

The stemflow infiltration experiments generally showed a bimodal distribution in the location of blue dye stains with depth, with a higher prominence of dye deeper in the soil. This suggests that infiltration of stemflow increases soil moisture at depth more than at the surface. Stemflow bypassed surface soil layers via preferential flow pathways along roots, which is consistent with the results from Liang *et al.* (2011). However, in this study, it appears that stemflow also moved through the loose organic rich topsoil between roots. Blue dye was visible along fine root hairs and medium roots, as well as in the soil at the end of these roots, which is similar to the results of Bogner *et al.* (2010), who showed the interaction of stemflow in roots and the soil matrix.

Clay layers at approximately 40 and 70-cm below the soil surface caused much of the stemflow to remain above these depths and to flow laterally along it, particularly in SIE 1 (Figures 8 and 9). Similar results were observed by Schwärzel *et al.* (2012) in a sprinkling experiment on a European Beech (*Fagus grandifolia*) tree, where a lower hydraulic conductivity layer impeded the percolation of stemflow into the deeper soil layers. The impedance of the percolation of stemflow can induce a perched water table beneath the tree (Germer, 2013). In this study site, only a few roots grew through the clay layers, but where roots were able to grow through this layer, it was generally along these roots that stemflow infiltrated deeper into the soil. Blue dye was also found in small cracks in the clay layer, suggesting that stemflow was able to infiltrate into the impeding soil layer.

The throughfall infiltration experiment generally showed a steady decrease in dye presence with depth. This is consistent with other studies that looked at infiltration in forest soils (e.g. Bachmair *et al.*, 2009; Bogner *et al.*, 2010). Throughfall also infiltrated deep into the soil along live and dead roots. This is similar to the results of Liang *et al.* (2011), who showed that throughfall flowed through the same preferential flow pathways as stemflow, but that it took longer for throughfall to flow to depth, and that it did not infiltrate as deep as stemflow (Liang *et al.*, 2011). The observed differences in the maximum depth of the blue dye stains for SIE 1, SIE 2, and TIE (122, 77, and 85 cm respectively) appear to indicate that stemflow infiltrated deeper into the soil than throughfall. However, when these depths are compared with the depth to bedrock, it appears that the dye infiltrated generally to the same depth above the bedrock (9, 14, and 6 cm respectively). Although blue dye was not visible on the bedrock, subsurface flow along the bedrock was observed during both stemflow excavations. It is possible that the blue dye that reached this layer was diluted or washed away by subsurface flow (~5–10 cm above the bedrock) during the rainfall events that occurred during the 4 to 5 weeks of each excavation. Small differences in the simulated storm size may have had a small influence on the maximum infiltration depth as well. The storm size for TIE was approximately 10 mm larger than the storm size for SIE 1 and 2 and may have increased the maximum infiltration depth for the TIE. Despite the varying soil depth (and storm size), it is likely that stemflow increased the maximum infiltration depth. The larger distribution of roots in the soil beneath the trees compared with the distribution of roots in the throughfall plot and the bimodal distribution of the blue dye for the stemflow experiments suggest that precipitation infiltrating as stemflow has a greater probability of flowing along roots to depth. Furthermore, the high stemflow intensities observed in this study suggest a faster transport of water to the soil–bedrock interface or the clay layer, particularly during periods of higher precipitation intensity later in the event, and therefore likely results in faster subsurface stormflow (Crabtree and Trudgill, 1985) or enhanced bedrock groundwater-recharge (Taniguchi *et al.*, 1996). The high-peak stemflow intensities suggest that more stemflow can reach the dense clay layer and enhance lateral subsurface stormflow than throughfall. The higher rate may exceed the infiltration capacity of the clay layer and result in enhanced lateral flow. Although lateral flow during the experiments resulted in stemflow flowing away from the trees, it did not reach the stream. This is more likely to occur during real events when rainfall occurs over the entire forest and not just a single tree, and during wetter conditions when preferential flow pathways may connect to each other and the stream (Noguchi *et al.*, 1999; Sidle *et al.*, 2001).

Stemflow preferentially flowed to the downslope side of the tree for SIE 1, similar to the results of Liang *et al.* (2011) and Schwärzel *et al.* (2012), but the upslope and downslope differences were minimal for SIE 2. This highlights the need to study preferential flow patterns around multiple trees, despite the very significant time and labour costs for each manual excavation.

The stemflow infiltration experiments were conducted on relatively large trees (29 and 52 cm DBH), while the stemflow funnelling ratios suggest that smaller trees may funnel more stemflow per unit basal area than larger trees. Blue dye experiments should therefore also be performed on smaller trees to determine if lateral flow and fast recharge to depth may be larger for small trees and because these trees have a different root structure and distribution.

CONCLUSION

Stemflow was only 1% of gross incident precipitation in the studied mature coastal coniferous forest. However, the funnelling ratios for the smaller trees indicated that stemflow can concentrate precipitation at the tree bole by up to 20 times. Stemflow intensities increased throughout the event and were generally largest for periods of higher rainfall intensity later in the event. The peak stemflow intensities were (much) higher than the peak precipitation intensities.

Stemflow was contained within 50–100 cm of the tree and generally flowed through preferential flow pathways around roots. There was a bimodal distribution in the location of blue dye stains with depth below the soil surface, with a higher prominence of dye deeper in the soil. Throughfall, on the other hand, flowed primarily through the soil matrix and along (fine) roots. Although stemflow appeared to infiltrate deeper into the soil (122 cm) than throughfall (85 cm), there was little difference in the infiltration depth relative to the soil depth. As a result, it is not clear if double funnelling of stemflow leads to deeper infiltration than throughfall. Lateral flow of stemflow was observed above a dense clay layer and along tree roots. The preferential flow pathways may allow stemflow to bypass the soil matrix and cause a larger increase in soil moisture at depth than at the surface. Thus, despite the negligible contribution of stemflow to the water balance, stemflow may have a significant effect on the hydrological processes in a mature coastal forest.

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